Variance Estimation for Sensor Data in Kalman Filters

A Practical Guide for UTS/IMU Fusion

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Contents

1 Introduction

When fusing sensor data in a Kalman filter (KF), the measurement noise covariance matrix $\mathbf{R}$ characterizes the variance of the sensor noise. Accurate estimation of $\mathbf{R}$ is critical:

- **Too small R**: Filter over-trusts noisy measurements, causing jitter.
- **Too large R**: Filter ignores measurements, relying excessively on prediction.

This document explores the challenges of estimating sensor variance—specifically for a Universal Total Station (UTS) at 10 Hz fused with an IMU at 50 Hz—and compares several estimation approaches.

2 Sample Size for Population Variance

2.1 Confidence Interval for Variance

For n i.i.d. samples from a normal distribution with true variance σ^2, the sample variance s^2 follows a scaled chi-squared distribution:

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1} \tag{1}$$

The $100(1-\alpha)\%$ confidence interval for σ^2 is:

$$\left[\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}},\quad \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}\right] \tag{2}$$

2.2 Required Sample Sizes (95% CI)

Desired precision (±% of true variance)	Approx. sample size
±20%	~ 50
±10%	~ 200
±5%	~ 800

Table 1: Sample sizes for population variance estimation at 95% confidence.

2.3 Effect of Autocorrelation

Sensor data at constant frequency is typically autocorrelated, reducing the effective sample size:

$$n_{\text{eff}} = \frac{n}{1 + 2\displaystyle\sum_{k=1}^{M} \rho(k)} \tag{3}$$

where $\rho(k)$ is the autocorrelation at lag k. For a first-order approximation with dominant lag-1 autocorrelation ρ:

$$n_{\text{eff}} \approx \frac{n}{1 + 2\,\dfrac{\rho}{1-\rho}} = \frac{n(1-\rho)}{1+\rho} \tag{4}$$

Raw samples	Duration @ 10 Hz	n_{eff} ($\rho = 0.9$)	Variance precision (95% CI)
50	5 s	~ 3	Very poor
100	10 s	~ 5	±60–80%
200	20 s	~ 11	±40–50%
500	50 s	~ 26	±25–30%

Table 2: Effective sample sizes for UTS at 10 Hz with high autocorrelation.

3 Approaches to R Estimation

3.1 Overview of Methods

1. **Offline characterization**: Collect stationary data, compute variance, hardcode into **R**.
2. **Manufacturer specification**: Use stated accuracy spec $\Rightarrow \sigma^2$.
3. **Online raw sample variance**: Rolling window of raw measurements. Fails under motion.
4. **Online with moving-mean subtraction**: High-pass filter before variance. Partial fix.
5. **Innovation-based adaptive**: Use KF innovation sequence. Recommended.

3.2 Recommended: Static R with Innovation Monitoring

For most production systems:

- Hardcode **R** from offline characterization or spec sheet.
- Monitor innovations online for anomaly detection (multipath, obstruction).
- The KF is **robust to moderate R misspecification**—being off by 2× typically causes only slight performance degradation.

4 Why Raw Sample Variance Fails Under Motion

4.1 The Fundamental Problem

The sample variance:

$$s^2 = \frac{1}{N-1} \sum_{i=1}^{N} (z_i - \bar{z})^2 \tag{5}$$

captures **both** measurement noise **and** any deterministic position change within the window. Under motion, the position change dwarfs the noise.

4.2 Analysis: $N = 5$ Samples (0.5 s @ 10 Hz)

Assume true UTS noise $\sigma = 3\,\text{mm}$, so $\sigma^2 = 9 \times 10^{-6}\,\text{m}^2$.

4.2.1 Case 1: Stationary

$$\text{True positions:} \quad \{0, 0, 0, 0, 0\}\ \text{m} \tag{6}$$

$$\text{Motion variance:} \quad 0 \tag{7}$$

$$\text{Expected } s^2 = \sigma^2 = 9 \times 10^{-6}\ \text{m}^2 \quad \checkmark \tag{8}$$

Result: Correct, but wide 95% CI due to only 4 degrees of freedom.

4.2.2 Case 2: Constant Velocity — 1 m/s

$$\text{True positions:}\quad \{0,\ 0.1,\ 0.2,\ 0.3,\ 0.4\}\text{ m} \tag{9}$$
$$\bar{p} = 0.2\text{ m} \tag{10}$$

Motion variance:

$$\text{Var}_{\text{motion}} = \frac{1}{4}\left[(0.04 + 0.01 + 0 + 0.01 + 0.04)\right] = 0.025\text{ m}^2 \tag{11}$$

$$\text{Expected } s^2 = \sigma^2 + 0.025 \approx 0.025\text{ m}^2 \tag{12}$$

Result: Overestimate by $\sim$ **2,800×**.

4.2.3 Case 3: Acceleration — 1 m/s² from Rest

$$\text{True positions } \left(\tfrac{1}{2}at^2\right):\quad \{0,\ 0.005,\ 0.02,\ 0.045,\ 0.08\}\text{ m} \tag{13}$$
$$\bar{p} = 0.03\text{ m} \tag{14}$$

Motion variance:

$$\text{Var}_{\text{motion}} = \frac{1}{4}\sum_{i=0}^{4}(p_i - 0.03)^2 \approx 1.06\times 10^{-3}\text{ m}^2 \tag{15}$$

Result: Overestimate by $\sim$ **119×**.

4.3 Analysis: $N = 25$ Samples (2.5 s @ 10 Hz)

Larger window makes things **worse** under motion:

4.3.1 Constant Velocity — 1 m/s

$$\text{True positions:}\quad \{0,\ 0.1,\ 0.2,\ \ldots,\ 2.4\}\text{ m} \tag{16}$$
$$\bar{p} = 1.2\text{ m} \tag{17}$$

$$\text{Var}_{\text{motion}} = \frac{0.01}{24}\sum_{i=0}^{24}(i-12)^2 = \frac{0.01\times 1250}{24} = 0.521\text{ m}^2 \tag{18}$$

Result: Overestimate by $\sim$ **57,900×** — 20× worse than $N = 5$.

4.3.2 Acceleration — 1 m/s²

$$\text{Var}_{\text{motion}} \approx 0.427\text{ m}^2 \tag{19}$$

Result: Overestimate by $\sim$ **47,400×** — $\sim$ 400× worse than $N = 5$.

4.4 Summary: Raw Variance Under Motion

Scenario	$N = 5$ **(0.5 s)**		$N = 25$ **(2.5 s)**	
	s^2	Overest.	s^2	Overest.
Stationary	9×10^{-6}	1×	9×10^{-6}	1×
1 m/s const. vel.	0.025	2,800×	0.521	57,900×
1 m/s^2 accel.	1.07×10^{-3}	119×	0.427	47,400×

Table 3: Raw sample variance is catastrophically inflated under motion.

Key insight: Motion variance grows as $\sim v^2T^2$ (velocity) or $\sim a^2T^4$ (acceleration), completely dominating noise for any practical window size.

5 Subtracting the Mean vs. Detrending vs. Innovation

5.1 What Does Subtracting the Mean Do?

A common misconception: "If I subtract the mean before computing variance, will it remove motion?"

No. The sample variance **already** subtracts the mean:

$$s^2 = \frac{1}{N-1} \sum (z_i - \bar{z})^2 \tag{20}$$

Subtracting the mean only removes a **constant offset**—it does nothing about trends within the window.

5.2 What Each Approach Removes

What you subtract	**Removes**	**Still contaminated by**
Mean (= sample variance)	DC offset	Linear trend, acceleration, any dynamics
Linear fit (detrending)	DC offset + constant velocity	Acceleration, jerk
Quadratic fit	DC offset + velocity + acceleration	Jerk, higher-order dynamics
KF prediction $H\hat{\mathbf{x}}_k^-$	Full predicted state at **each** timestep	Only true noise (by design)

Table 4: Hierarchy of motion removal approaches.

5.3 Example: 1 m/s motion, $N = 5$

$$\text{Raw data:} \quad \{0.0,\ 0.1,\ 0.2,\ 0.3,\ 0.4\} \qquad \text{(dominated by motion)} \tag{21}$$

$$\text{Subtract mean:} \quad \{-0.2,\ -0.1,\ 0.0,\ 0.1,\ 0.2\} \qquad \text{(still a ramp, } s^2 \approx 0.025\text{)} \tag{22}$$

$$\text{Subtract linear trend:} \quad \{\sim 0,\ \sim 0,\ \sim 0,\ \sim 0,\ \sim 0\} \qquad \text{(only noise remains)} \tag{23}$$

$$\text{Subtract KF prediction:} \quad \{\varepsilon_1,\ \varepsilon_2,\ \varepsilon_3,\ \varepsilon_4,\ \varepsilon_5\} \qquad \text{(only noise remains)} \tag{24}$$

6 Moving-Mean Subtraction Approach

6.1 Method

Compute a moving mean, subtract it from the signal, then compute variance on the residual:

$$\tilde{z}_k = z_k - \bar{z}_k^{(M)}, \qquad \bar{z}_k^{(M)} = \frac{1}{M}\sum_{i=0}^{M-1} z_{k-i} \tag{25}$$

This is essentially a **high-pass filter**.

6.2 Constant Velocity Analysis

For a ramp $p_k = v \cdot k\Delta t$, the moving mean has a systematic lag:

$$\bar{z}_k^{(M)} \approx p_k - v\Delta t \cdot \frac{M-1}{2} \tag{26}$$

The residual:

$$\tilde{z}_k = \underbrace{v\Delta t \cdot \frac{M-1}{2}}_{\text{constant bias}} + \underbrace{\epsilon_k - \frac{1}{M}\sum_{i=0}^{M-1} \epsilon_{k-i}}_{\text{high-pass filtered noise}} \tag{27}$$

The constant bias **does not affect variance**. The noise term has variance:

$$\mathrm{Var}(\tilde{z}_k) = \sigma^2\left(1 - \frac{1}{M}\right) = \sigma^2 \cdot \frac{M-1}{M} \tag{28}$$

Result for constant velocity: This approach works, with a slight underestimate by factor $\frac{M-1}{M}$.

6.3 Acceleration Analysis

For $p_k = \frac{1}{2}a(k\Delta t)^2$, the moving mean cannot track a parabola. The residual contains a **time-varying bias**:

$$\mathrm{bias}_k \propto a \cdot \Delta t^2 \cdot k \cdot \frac{M-1}{2} \tag{29}$$

Since this bias **changes with time**, it inflates the variance window.
For $a = 1\,\mathrm{m/s^2}$, $\Delta t = 0.1\,\mathrm{s}$, $M = 5$, $N = 5$:

$$\text{Overestimate} \approx 5\times \tag{30}$$

Better than raw (119×), but still biased.

6.4 Autocorrelation Issue

The residuals $\tilde{z}_k = \epsilon_k - \bar{\epsilon}_k^{(M)}$ are **autocorrelated** because adjacent samples share overlapping mean windows. This means:

- Fewer effective degrees of freedom than expected.
- Larger N needed for reliable estimates.
- Variance is biased by factor $\frac{M-1}{M}$.

6.5 Summary: Moving-Mean Approach

Scenario	**Raw var.**	**Moving-mean sub.**	**Innovation-based**
Stationary	σ^2 (correct)	$\frac{M-1}{M}\sigma^2$ (slight underest.)	σ^2 (correct)
1 m/s const. vel.	2,800× overest.	$\sim$ 1× (works)	1× (correct)
1 m/s^2 accel.	119× overest.	$\sim$ 5× overest.	1× (correct)
Jerk / complex motion	Catastrophic	Still leaks	1× (correct)

Table 5: Comparison of variance estimation methods under different motion profiles.

7 Innovation-Based Adaptive Estimation

7.1 The Innovation Sequence

The KF innovation (measurement residual) at time step k is:

$$\boldsymbol{\nu}_k = \mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_k^- \tag{31}$$

where $\hat{\mathbf{x}}_k^-$ is the **predicted state** (prior). This subtraction removes the predicted motion—including velocity, acceleration, and any dynamics captured by the state model—leaving only the measurement noise.

7.2 Adaptive R Estimation

Estimate $\mathbf{R}$ from the last N innovations:

$$\boxed{\hat{\mathbf{R}}_k = \frac{1}{N}\sum_{i=k-N+1}^{k} \boldsymbol{\nu}_i\boldsymbol{\nu}_i^T - \mathbf{H}\mathbf{P}_k^-\mathbf{H}^T} \tag{32}$$

The term $\mathbf{H}\mathbf{P}_k^-\mathbf{H}^T$ corrects for the prediction uncertainty contribution to the innovation covariance.

7.3 Properties

- **Handles arbitrary dynamics**: Everything captured by the state model is removed.
- **Minimal latency**: $N = 15$–20 samples (1.5–2 s at 10 Hz) is sufficient.
- **No extra computation**: The innovation $\boldsymbol{\nu}_k$ is **already computed** in every KF update step.
- **No autocorrelation issue**: Innovations of a correctly tuned KF are white (uncorrelated).
- **Self-consistent**: Uses the filter's own prediction model rather than an external assumption.

7.4 Recommended Configuration for UTS (10 Hz) + IMU (50 Hz)

Parameter	**Recommendation**
Primary approach	Hardcode $\mathbf{R}$ from offline characterization or spec sheet
Adaptive window N	15–20 samples (∼2 s at 10 Hz)
Cold start	Use conservative (larger) $\mathbf{R}$ for first 2–3 s
Monitoring	Check innovation consistency for anomaly detection

Table 6: Production configuration for UTS/IMU Kalman filter.

8 Summary of Key Takeaways

1. **Raw sample variance is only valid when stationary.** Under any motion, it produces massively inflated estimates because deterministic position changes are aliased as noise.
2. **Larger windows make it worse under motion.** Motion variance grows as $\sim v^2T^2$ (velocity) or $\sim a^2T^4$ (acceleration).
3. **Subtracting the mean does not help.** Sample variance already subtracts the mean. A ramp minus its mean is still a ramp.
4. **Moving-mean subtraction partially helps** for constant-velocity motion but fails for acceleration and introduces autocorrelation in the residuals.
5. **Innovation-based estimation is the principled solution.** It removes all predicted dynamics, requires no extra computation, and works for arbitrary motion profiles.
6. **For production: hardcode R, monitor innovations.** The KF is robust to moderate $\mathbf{R}$ misspecification. Use innovation monitoring for anomaly detection rather than continuous $\mathbf{R}$ adaptation.

Quick Reference

Innovation:	$\boldsymbol{\nu}_k = \mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_k^-$
Adaptive $\mathbf{R}$:	$\hat{\mathbf{R}}_k = \dfrac{1}{N}\sum_{i=k-N+1}^{k} \boldsymbol{\nu}_i\boldsymbol{\nu}_i^T - \mathbf{H}\mathbf{P}_k^-\mathbf{H}^T$
Window size:	$N = 15\text{–}20 \ \ (\sim 2\,\mathrm{s}\text{ at }10\,\mathrm{Hz})$
Moving-mean bias:	$\mathrm{Var}_{\mathrm{est}} = \sigma^2 \cdot \dfrac{M-1}{M}$
Effective samples:	$n_{\mathrm{eff}} = \dfrac{n(1-\rho)}{1+\rho}$